CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 24: Solution of Right Triangles

Every formula and result of the chapter on one sheet

1	Perpendicular	Hypotenuse $\times \sin\theta$
2	Base	Hypotenuse $\times \cos\theta$
3	Perpendicular from the base	Base $\times \tan\theta$
4	Hypotenuse	$\frac{\text{perpendicular}}{\sin\theta} = \frac{\text{base}}{\cos\theta}$
5	Acute angles	$\angle A + \angle C = 90^\circ$
6	30-60-90 triangle	$1 : \sqrt{3} : 2$
7	45-45-90 triangle	$1 : 1 : \sqrt{2}$
8	Side opposite 30°	Half the hypotenuse
9	Diagonal of a square	Side $\times \sqrt{2}$
10	Altitude of an equilateral triangle	$\frac{\sqrt{3}}{2} \times$ side
11	Chord subtending 2θ at the centre	$2r \sin\theta$
12	Surd values	$\sqrt{2} \approx 1.414, \sqrt{3} \approx 1.732$